

Learning Sub-Grid Physics of Geophysical Turbulence with Fourier Neural Operators: A Case Study in Quasi-Biennial Oscillation Modelling

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Abstract

Eddy parameterization is known to play a fundamental role in earth system models. The Fourier Neural Operator (FNO) is a newly-proposed deep learning algorithm specialized in learning PDEs. Here we use the gravity wave parameterization in the 1D-QBO model as a testbed to test the ability of FNOs in eddy parameterization. An FNO with a very small parameter size can simulate the QBO period more accurately than a deep MLP in this task. However, FNOs still have difficulty learning the wave forcing with a wide wave spectrum. These findings demonstrate the potential of FNOs over traditional MLPs in gravity wave parameterization of QBO and provide a possible way for integrating AI-based schemes into complex global circulation models.

1 Introduction

Ocean and climate models simulate the dynamics of the geophysical flow by solving the associated differential equations numerically. For computational representation, the continuous fields of physical variables (e.g., velocity, pressure, temperature) in the atmosphere and ocean must be discretized. The most intuitive approach is to divide the fluid into a number of grid cells and approximate the continuous field by the average value across the grid cell or the value at the center of the grid cell (Neelin [2010]). This can approximately capture the behavior of motions at spacial scales much larger than the grid cell, but obviously omits the other values at different points within the grid cell. Therefore, higher resolution models enable more detailed simulations and explicitly captures the complexities of physical systems. However, if we try to increase the spatial resolution, the consumption of computation cost will grow exponentially. Additionally, according to the CFL condition, the time step should be shortened correspondingly to maintain numerical stability (De Moura and Kubrusly [2013]). Thus, there is a tradeoff between physical accuracy and computing cost.

To bridge the aforementioned gap, one strategy is to employ a coarse resolution model and construct functions of resolved variables to parameterize the processes whose spatial scales are smaller than the grid cell, which are called subgrid-scale parameterizations (Uliasz [1994]). Since geophysical turbulence is inherently multi-scale (McWilliams [1989]), among different types of subgrid parameterization, eddy parameterization has consistently remained a critical topic in both ocean and climate models (Fox-Kemper et al. [2019]; Bauer et al. [2015]). Gravity waves (GWs) are one type of fast waves in the geophysical fluid system. Considering their high frequency and short wavelength, the computational cost will be very high in order to model them explicitly. However, gravity waves are the momentum sources for the large-scale motions in the atmosphere, thus affecting the long-term climate predictions significantly. Therefore, Gravity Wave Parameterization (GWP) is one important component in numerical models.

A prominent example of GW-driven dynamics is the Quasi-Biennial Oscillation (QBO), which is the dominant large-scale interannual variability in the tropical stratosphere. It is characterized by the downward-propagating reversals of the mean zonal winds, with a period of approximately 28 months (Baldwin et al. [2001]). The mechanism of QBO can be explained by the absorption of GWs at critical layers where the mean-flow speed is close to the phase speeds of GWs (Plumb [1977]). Realistic GWP are crucial for simulating the QBO in Global Circulation Models (GCMs, Richter et al. [2020]), and consequently affect other signals in the atmosphere (such as MJO and subtropical jets) through

teleconnections, which are closely related to the subseasonal-to-seasonal forecast skills (Anstey et al. [2022a]). Garfinkel et al. [2022]’s study has shown that while the QBO’s period and amplitude can be effectively tuned through parameterized gravity waves, its vertical structure remains poorly captured and difficult to simulate accurately in the GCMs. Thus, we adopt the QBO as a testbed to study the general geophysical turbulence parameterization.

Over the past decade, Machine Learning (ML) methods have been increasingly applied to the eddy parameterization in the atmospheric and oceanic sciences (de Burgh-Day and Leeuwenburg [2023]). Deep neural networks have demonstrated their potential to parameterize convection processes in climate models (Rasp et al. [2018]) and 2D turbulence in ocean models (Yan et al. [2025]). The current deficiencies of ML-driven parameterization primarily lie in the physical inconsistency and their generalization ability. ML models cannot learn the conservation laws directly from training data alone thus and need extra physical constraints (Beucler et al. [2021]). Regarding generalization, O’Gorman and Dwyer [2018]’s study also found that global warming can be accurately simulated only when training samples from both the control and warm climate are used. Existing ML strategies for GWP in QBO mainly focus on convolutional neural networks, whose performances are sensitive to the receptive field (Pahlavan et al. [2025]). Recently, Fourier Neural Operator (FNO) was proposed as an operator learning method and has shown strong capability in modeling the Navier–Stokes equations and other differential equations-based problems (Li et al. [2020]). These results suggest FNOs have potential for sub-grid turbulence parameterization, which is also characterized by intense nonlinearity and complex interactions. Therefore, in this study, we will investigate the capability of FNO for GWP within a 1D-QBO testbed. In Section 2, we introduce the 1D-QBO model we employ and the model architecture of FNO in detail. Section 3 describes our results, and the discussion and summary are provided in Section 4.

2 Methods

2.1 1D-QBO Model

The 1D-QBO model, introduced by Plumb [1977], represents a one-dimensional prototype of the tropical stratosphere. It involves a source of GWs from the lower boundary, which is derived from a Boussinesq system based on WKB approximation (Plumb [1975]). In this study, the model is expressed as

$$\frac{\partial \bar{u}}{\partial t} + w \frac{\partial \bar{u}}{\partial z} - \kappa \frac{\partial^2 \bar{u}}{\partial z^2} = \text{GWF}(\bar{u}, z), \quad (1)$$

where,

$$\text{GWF}(\bar{u}, z) = -\frac{\rho_L}{\rho_0} \sum_n \frac{\partial F_n}{\partial z}; \quad F_n = F_L \exp \left\{ -\int_{z_L}^z \frac{N\mu}{k_n[\bar{u}(z) - c_n]^2} dz' \right\}. \quad (2)$$

The n denotes the number of wave modes and k_n, c_n are the wavenumber and wave phase speed respectively. The subscript L denotes the variables at the bottom of stratosphere. N here is the buoyancy frequency and μ is thermal dissipation rate of waves. Although the 1D-QBO model neglects the Coriolis effect, it provides a fundamental framework for simulating QBO-like wind fields with downward propagating phases. Plumb [1977] found that only two opposite wave modes are required to produce a QBO phase if thermal dispersion exists. Figure 1 shows the numerical solution of the 1D-QBO model with the boundary conditions,

$$\bar{u}(z = 17 \text{ km}) = \bar{u}(z = 35 \text{ km}) = 0 \text{ m s}^{-1}. \quad (3)$$

The parameters in Equation 2 are set to $n = 2$ and $c_1 = -c_2 = 32 \text{ m} \cdot \text{s}^{-1}$, following Plumb [1977]. We employ this numerical solution as the training dataset for FNO models to learn. Other basic parameters of the 1D-QBO model are shown in the Table 1.

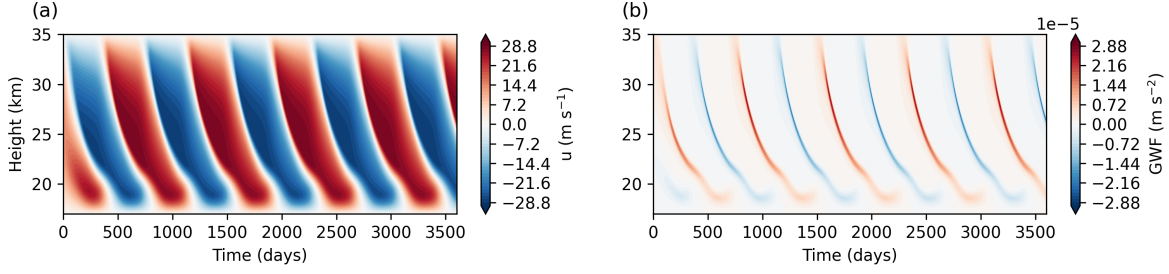


Figure 1: The numerical solution of the 1D-QBO model. (a) the zonal wind velocity $\bar{u}$; (b) the gravity wave forcing GWF.

Parameters	Quantity
buoyancy frequency N	$2.16 \times 10^{-2} \text{ s}^{-1}$
diffusion rate κ	$0.3 \text{ m}^2 \text{ s}^{-1}$
n	2
c_1	32 m s^{-1}
c_2	-32 m s^{-1}
wavenumber k	$2\pi/40000 \text{ km}^{-1}$
wave dissipation rate μ	$1.2 \times 10^{-6} \text{ s}^{-1}$

Table 1: Parameters for 1D-QBO model.

2.2 Fourier Neural Operator

Fourier Neural Operator is a kind of Neural Operator which is designed to solve PDEs. Instead of learning maps from $\mathbb{R}^n \rightarrow \mathbb{R}^m$ like most traditional neural networks, neural operators aim to learning maps in function spaces, i.e. learning the operator

$$\mathcal{G} : v(x) \rightarrow u(x), \quad (4)$$

where $v(x)$ is the input and $u(x)$ is the output (Kovachki et al. [2023]). In a classic PDE problem, $v(x)$ can be initial conditions, boundary conditions or forcing and $u(x)$ is the solution. Thus, the neural operators are resolution-invariant because a function can be represented under different sampling schemes.

In order to guarantee the differentiability, The framework of neural operators is similar to neural networks,

$$u = (K_l \circ \sigma_l \circ \dots \circ \sigma_1 \circ K_1)(v), \quad (5)$$

where K_i are linear transforms and σ_i are activation functions. For FNO, K_i are Fourier layers, containing a Fourier transform and an inverse Fourier transform. The structure of a Fourier layer is demonstrated in Figure 2 (Li et al. [2020]). The input function $v(x)$ is first transformed into the frequency domain by a Fourier transform. A linear transformation R is then applied to the Fourier coefficients. The linear transforms in Fourier space are convolutions in real space, so FNOs resemble CNNs but with full domain extent, compared to the finite kernel size in CNNs. In practice, the spectrum is always truncated to avoid high frequency oscillation, which means the corresponding rows (or columns) of matrix R are set to zero. After that, the result $R\mathcal{F}(x)$ is mapped back to the original space through an inverse Fourier transform. Finally, following the idea of residual connections, a linear transformation of the input function, $Wv(x)$, is added to the output of the Fourier layer. Since the derivatives of the Fourier transform and its inversion can both be expressed analytically, the Fourier layer is differentiable. The overall architecture of the fourier layer can be written as,

$$k(v) = \mathcal{F}^{-1}(R\mathcal{F}(v)) + Wv, \quad (6)$$

where R and W are the learnable parameters.

The Fourier layer is implemented using PyTorch to take advantage of its automatic differentiation capabilities. `torch.fft.rfft` and `torch.fft.irfft` are utilized to efficiently transition between

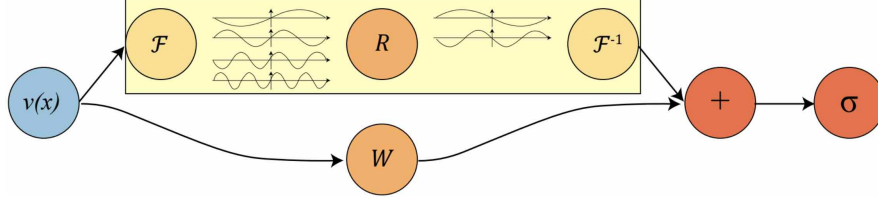


Figure 2: The architecture of the Fourier layer (Li et al. [2020], Figure 2b).

physical and spectral domains. The linear transform in the spectral space is realized by the function `compl_mulld`, which enables complex multiplication after mode truncation. PyTorch integrates the above operations into a computational graph and records all the local gradients as the data go through these functions. Therefore, the gradients are able to be backpropagated to every parameter if we call `loss.backward()`. Regarding the mode truncation, which is typically indifferentiable, the gradients associated with the unselected components are automatically assigned a value of zero.

```

1 class SpectralConv1d(nn.Module):
2     def __init__(self, in_channels: int, out_channels: int, modes: int):
3         super(SpectralConv1d, self).__init__()
4
5         self.in_channels = in_channels
6         self.out_channels = out_channels
7         self.modes = modes
8         self.scale = (1 / (in_channels*out_channels))
9         self.weights = nn.Parameter(self.scale * torch.rand(in_channels, out_channels,
10                                                                self.modes, dtype=torch.cfloat))
11
12     # Complex multiplication
13     def compl_mulld(self, input, weights):
14         return torch.einsum("bix,iox->box", input, weights)
15
16     def forward(self, x: torch.Tensor) -> torch.Tensor:
17
18         batchsize = x.shape[0]
19         # Fourier transformation
20         x_ft = torch.fft.rfft(x)
21
22         # Multiply relevant Fourier modes
23         out_ft = torch.zeros(batchsize, self.out_channels, x.size(-1)//2 + 1, device=
24                               x.device, dtype=torch.cfloat)
25         out_ft[:, :, :self.modes] = self.compl_mulld(x_ft[:, :, :self.modes], self.
26                                                       weights)
27
28         #Return to physical space
29         x = torch.fft.irfft(out_ft, n=x.size(-1))
30         return x

```

Listing 1: Implementation of Fourier layer in PyTorch.

2.3 Model Architecture

The architecture of the model is illustrated in Figure 3. The neural operator takes the zonal wind $\bar{u}^t$ at time t as input and predicts the corresponding GWF $\hat{f}^t$. First, the input is projected into a higher-dimensional feature space through a linear embedding layer. The dimension of this feature space is equal to `in_channels` of the Fourier layer in Listing 1. The tensor in feature space is then processed by a series of Fourier layers, each followed by an activation function. To put several Fourier layers in series, `in_channels` and `out_channels` are set to be identical. The output of the last Fourier layer is mapped to the target dimension through a fully connected layer.

The model is trained offline. The feature is $\bar{u}$ at every time step generated by the 1D-QBO model and the corresponding labels are given by $f^t = \text{GWF}(\bar{u}^t)$. These data are stochastically divided into

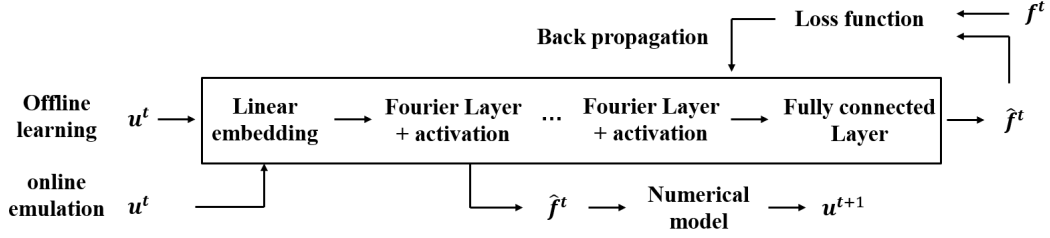


Figure 3: Model Architecture.

the training set and the validation set. The order in each set is also randomized. Considering that the training data are zero at the boundary, padding is not necessary in this model. The model performance is tested online. The predicted $\hat{f}^t$ is given to the numerical model to substitute $\text{GWF}(\bar{u}, z)$ in Equation 2. If the model is denoted by $\mathcal{M}$, the mathematical expression of online emulation is

$$\frac{\partial \bar{u}}{\partial t} + w \frac{\partial \bar{u}}{\partial z} - \kappa \frac{\partial^2 \bar{u}}{\partial z^2} = \mathcal{M}(\bar{u}). \quad (7)$$

To compare the performance between different models, we also consider a fully connected neural network, i.e., multilayer perceptron (MLP). The MLP consists of eight hidden layers in order to extract enough features from inputs. The parameters defined in the two models are shown in table 2. The setting of learning rates refers to Pahlavan et al. [2025]’s work, where the learning rate for FNO is much smaller than that for MLP. One potential explanation is that the FNO possesses fewer degrees of freedom and thus needs a more careful training session. In order to find a sufficiently capable competitor for FNO, The channels of MLP are chosen to be much larger than those of the FNO, as well as the parameter size.

	FNO	MLP
Number of layers	4	8
Learning rate	0.0001	0.1
Degrees of freedom	60012	2393978
Truncation mode	9	N/A
Channel	37	213 to 639

Table 2: Parameters for FNO and MLP.

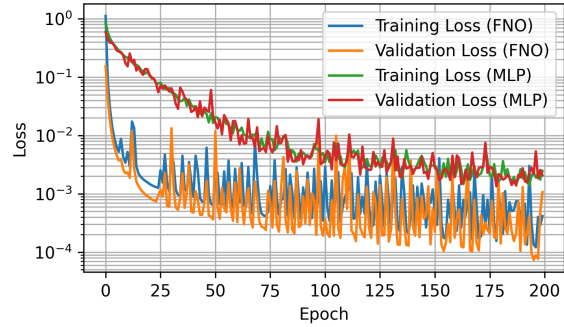


Figure 4: Training and validation loss with deterministic forcing data.

3 Result

3.1 Performance under Deterministic Training Data

We first consider a simple task in which the training datasets are generated without any noise. Both models were trained for 200 epochs. As shown in Figure 4, both models basically converged after 200 epochs. We still try smaller learning rate for MLP and the corresponding number of epochs, the results and conclusions won’t change significantly (not shown here). The online testing results of the FNO and MLP models are presented in Figure 5. The FNO model used here contains four Fourier layers. Both simulations are stable, consistent with the results in Pahlavan et al. [2025] because they take the full vertical profile as their receptive fields. However, the oscillation period produced by the MLP is shorter than that generated by the FNO: the MLP period is about 25.53 months, while FNO

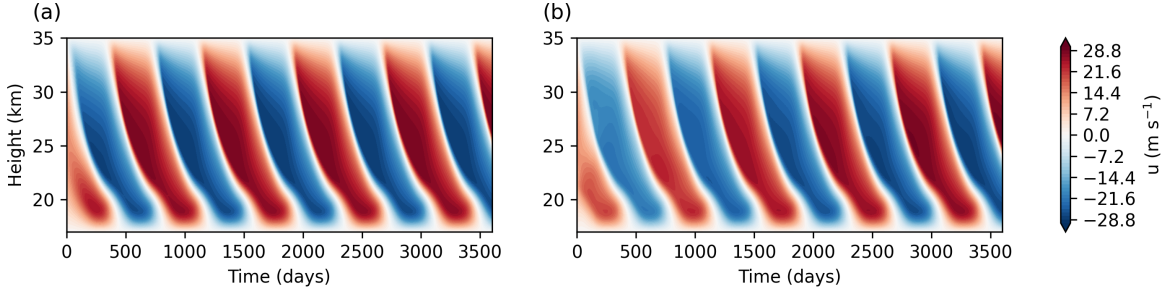


Figure 5: Online emulations (a) FNO, (b) MLP.

period is about 27.07 months, which is closer to the numerical model and real atmospheric phenomena.

Figure 6 demonstrates the offline and online GWF predictions provided by the two models. Although the two models perform excellently offline, the FNO model generates much smoother GWF, which contributes to its stable online performance. In contrast, the MLP predictions exhibit small fluctuations around $\text{GWF} \approx 0$. This instability exerts non-physical momentum flux onto the mean flow and thus causes the mean flow to be accelerated prematurely, which may shorten the oscillation period.

The improved period-stability of the FNO can possibly be understood from the structure of Fourier layers. As discussed in Section 2.2, the FNO retains only the first several low-wavenumber Fourier modes. This spectral truncation effectively suppresses the high-frequency fluctuations. In traditional neural networks (such as the MLP here), such fluctuations typically arise when the model encounter inputs that are not well represented in the training dataset. Specifically, when the true forcing is close to $\text{GWF} \approx 0$, the training data contain many exact or near-zero values. Since the models approximate this value through nonlinear combinations of parameters and inputs, small prediction errors occur when new inputs are presented, leading to non-physical oscillations around zero. By contrast, although the Fourier layer appears to impose a rigid restriction by retaining only a limited set of spectral modes, this constraint prevents small prediction errors from being directly projected onto the forcing field. This mechanism is similar to a low pass filter. Consequently, the predicted GWF remains close to zero when the GWF is weak, avoiding artificial perturbations to the mean-flow acceleration and possibly contributing to a more stable oscillation period.

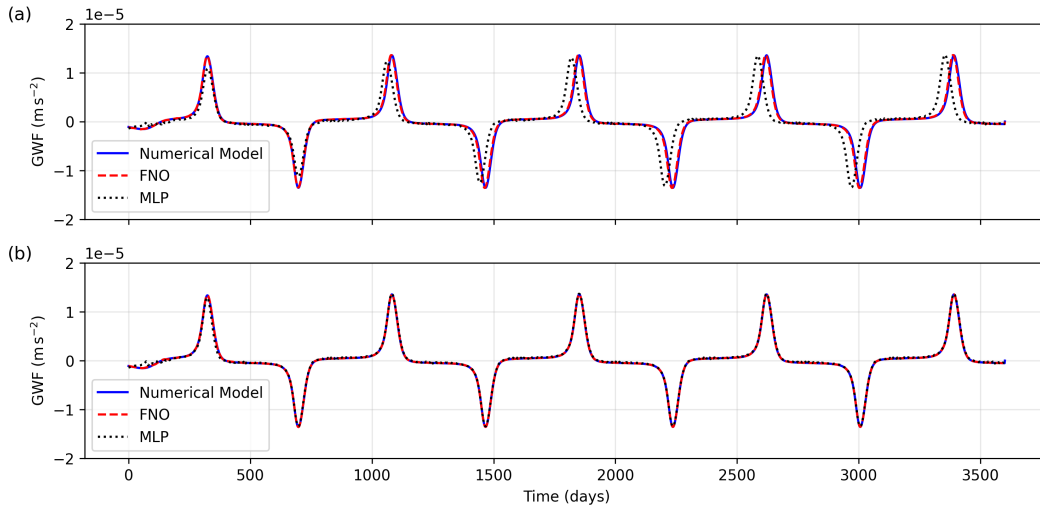


Figure 6: GWF prediction without noise (a) Online performances; (b) Offline performances.

3.2 Performance under Noisy Training Data

In the real atmosphere (or oceans), the eddy forcing acting on the mean flow is inherently noisy due to unresolved processes and broadband wave spectra (Berner et al. [2017]). This noise will affect the stability and accuracy of both numerical models (Hodyss et al. [2013]) and ML models (Wikner et al. [2024]). Therefore, it is essential to assess the performance and generalization ability of data-driven models when trained on noisy datasets. From a ML perspective, this is also a robustness test for the application of FNO in GWP.

In this study, two different approaches are adopted to incorporate noise into the training data. In the first experiment, training data are generated from the deterministic numerical model, and additive white noise is subsequently imposed on the resulting dataset. This setup allows for an assessment of the model robustness when learning a deterministic physical process. Figure 7 shows the corresponding results. The online performance demonstrates that the FNO still produces a stable period, while the period generated by the MLP is much shorter than that shown in Figure 6.

Second, stochasticity is introduced directly into the gravity-wave forcing by perturbing the phase speeds of the waves, which is more similar to the real stratosphere (Holton and Lindzen [1972]). The numerical model is then driven by this stochastic forcing to generate training data. Instead of focusing solely on the impact of noise on the training process from a ML perspective, this experiment aims to assess whether the models can capture the underlying physical mechanisms when stochasticity is introduced to the governing dynamics. In Figure 8, both the FNO and MLP fail to reproduce the exact oscillation period. Nevertheless, the FNO yields a period that is substantially closer to the reference.

The potential failure of the FNO in the second task may stem from this tendency to learn the numerical average of stochastic GWF. However, the actual underlying dynamical evolution is not a simple linear superposition of these forced states. For example, the dynamical response to a fast-wave forcing followed by a slow-wave forcing, or the reverse, differs significantly from the response to the simple average of these two forcing terms. This implies that offline training is not sufficient to capture the nonlinear effect of stochasticity on long-term evolution, thereby underscoring the importance of online training to maintain physical consistency in nonlinear systems (Pahlavan et al. [2024]). In online training, the loss function contains both neural networks and the numerical model.

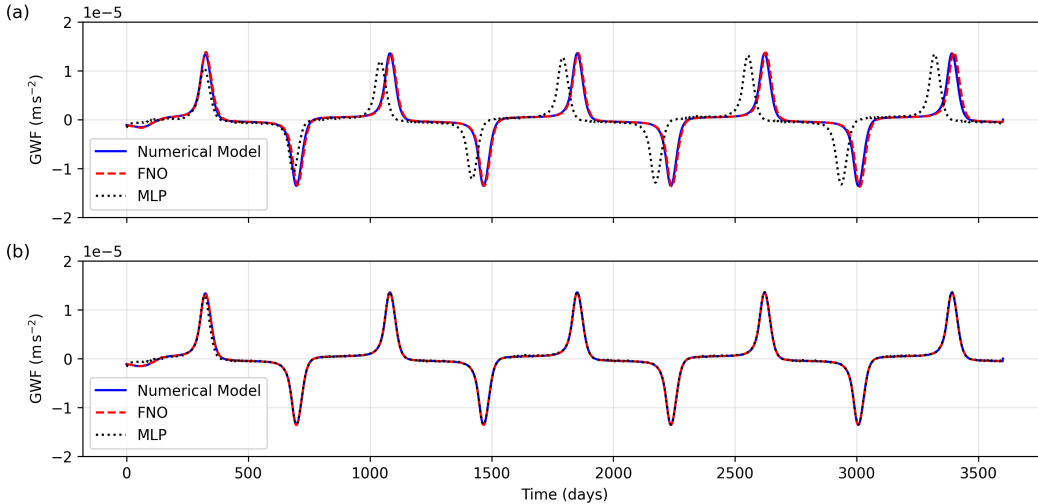


Figure 7: GWF prediction with noisy training data (a) Online performances; (b) Offline performances.

3.3 Performance of FNOs with Different Model Complexities

Although the above experiments demonstrate that the FNO is capable of learning subgrid physics in geophysical dynamics, its computational cost remains a practical concern. When trained on the

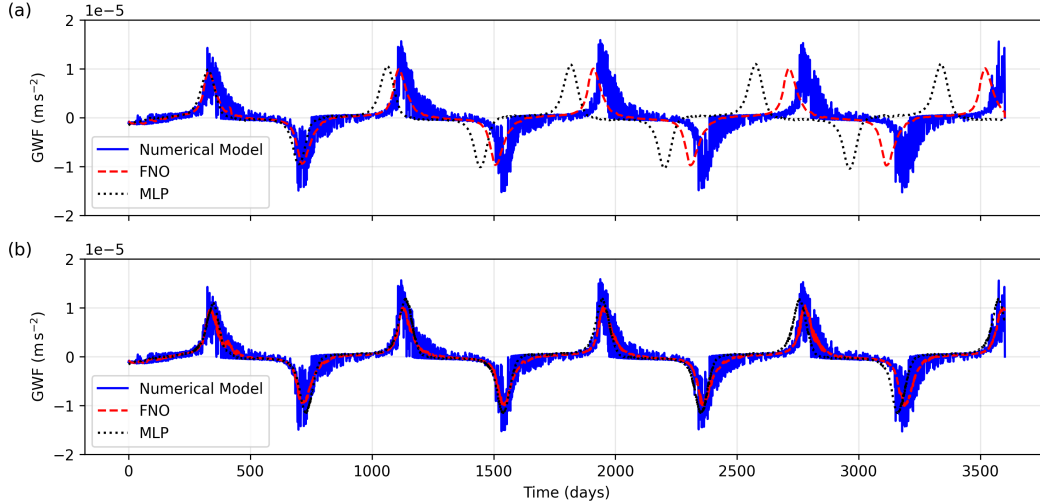


Figure 8: GWF prediction with stochastic model data (a) Online performances; (b) Offline performances.

CPU of a personal laptop, the FNO requires approximately 90–120 minutes, whereas the MLP completes training within 15–30 minutes under comparable settings.

To assess the tradeoff between model complexity and the performance, we examine FNO architectures with different numbers of Fourier layers. To eliminate the influence of stochastic noise for simplicity, we use deterministic forcing as a baseline. All models are trained for 200 epochs, which are sufficient to ensure convergence. The results indicate that models with insufficient complexity fail to produce stable online predictions. The FNO with one Fourier layer leads to rapid numerical divergence, while the one with two Fourier layers can generate a QBO-like structure with inconsistent periods at first, but then also becomes unstable eventually. Once the number of Fourier layers reaches three or higher (results with four layers can be seen in Section 3.1), the model is able to generate stable oscillations with a period of approximately 28 months.

In this section, we show that while the Fourier layer can extract the physical features in the QBO data, the model’s ability is still restricted by the total number of parameters and depth. More parameters and deeper architectures provide the model more degrees of freedom to tackle the unseen input and enhance stability (Belkin et al. [2019]). In this specific task, a minimum of three Fourier layers with more than 40000 parameters is required to achieve a stable climate state in online emulations, although that is still much lower than millions of parameters in the MLP.

4 Conclusions and Discussions

In this study, we briefly introduced the background of sub-grid scale turbulence parameterizations. The rapid development of AI models has demonstrated remarkable efficacy in capturing complex fluid dynamic processes. Among these, the Fourier Neural Operator (FNO), a novel architecture, has achieved superior performance in global weather forecasting. To evaluate the FNO’s capability in learning sub-grid physics, we employ a 1D Quasi-Biennial Oscillation (QBO) model as a testbed. The primary focus of this assessment is the model’s performance in online testing to see if it could faithfully reproduce the authentic QBO cycle. Previous studies and our findings all suggest that the offline accuracy of a model does not always translate to online stability. Although naive models can give relatively accurate results offline, their online performance is not necessarily unstable.

Under deterministic and noiseless conditions, the FNO demonstrates remarkable precision in maintaining the QBO period with noticeably fewer degrees of freedom compared to the MLP. While the MLP often suffers from significant phase drifts and subtle non-physical oscillations, the FNO success-

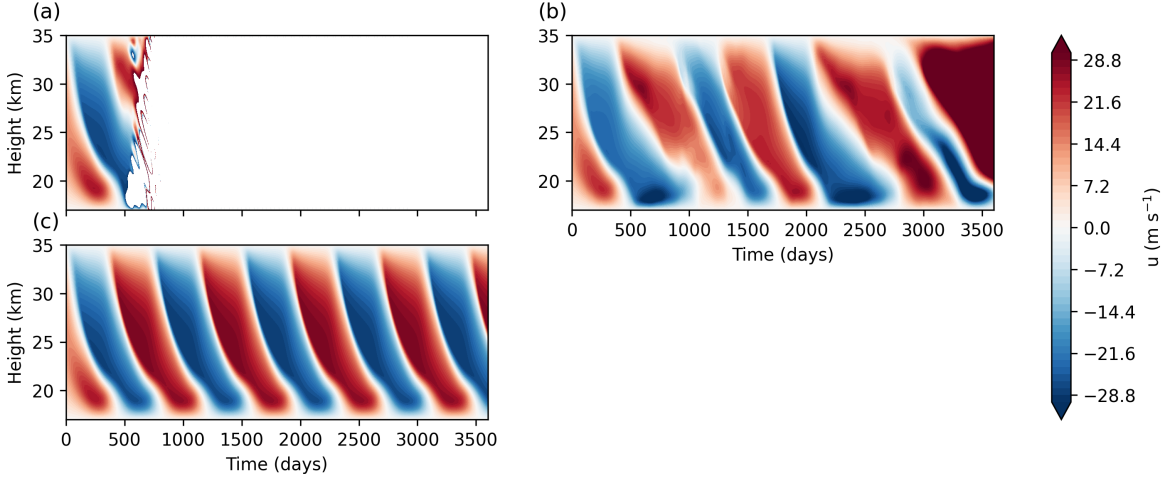


Figure 9: Online performances of the models with different complexity. (a) one Fourier Layer; (b) two Fourier Layers; (c) three Fourier Layers.

fully captures the sharp gradients associated with critical layer absorption. When additive white noise was introduced to the training data, the FNO exhibited robust generalization, whereas the MLP’s online performance degraded, resulting in a significantly shortened oscillation period. In experiments involving stochastic gravity-wave forcing, where the wave spectrum is broader, similar to that of the real stratosphere, both the FNO and the MLP fail to generate an accurate oscillation period, although the MLP’s period is much more inconsistent with the model truth. These results, on the one hand, illustrate the FNO shows enhanced robustness for this task, learning Gravity Wave Forcing (GWF) in 1D-QBO model. On the other hand, they imply that if we want to improve the generalization ability of the FNO, we need to make certain adjustments to the model structure, parameter size or training strategies. One potential strategy that has been widely used in eddy parameterization is online training, where the numerical model error is also taken into account in the training loss and thus the back propagation process.

Although the FNO can effectively learn the GWF in 1D-QBO model with a small set of parameters, the efficiency is another important indicator to evaluate the quality of a certain AI algorithm. Architectural sensitivity analysis revealed that the success of the FNO highly depends on its depth and complexity. We found that a minimum of three Fourier layers is essential to sustain a stable climate state in online emulations. With a parameter size of approximately 20,000, the FNO outperformed much larger MLP structures, further demonstrating its superiority to MLP architectures in gravity wave parameterization.

The outstanding performance of the FNO indicates that its architecture may in principle be able to discover the underlying physics in QBO. We notice the small fluctuation in the MLP’s prediction and contrast it with the smooth profile generated by the FNO. We qualitatively attribute this stability to the spectral truncation inherent in the FNO architecture. This mechanism effectively suppresses non-physical noise introduced by the neural networks/operators, thereby enhancing the physical consistency of the long-term emulation.

Despite these conclusions, the physical consistency of FNO-based parameterizations still need to be investigated quantitatively. In this study, we focus on the accuracy of QBO periods as an indicator. However, a critical question remains whether the neural networks/operators can obey the physical conservation laws and whether the superior performance of the FNO can be explained from this aspect. The violations of these natural rules have been widely found in AI weather prediction (Bonavita [2024]), and the performance was significantly enhanced by improving this aspect (Sha et al. [2025]). Besides, some works also computed the probability density function (PDF) of GWF prediction to examine the physical consistency (Pahlavan et al. [2024]). To improve the physical consistency and

preserve the advantages of neural operators, an alternative approach is Physics-Informed Neural Operator (PINO), which integrates data and physical constraints into the training session together (Li et al. [2024]). This strategy has been adopted in FNOs and shows better generalization ability in modeling chaotic Kolmogorov flows than traditional FNO and Physics-Informed Neural Networks (PINNs).

From a technical perspective, there are still many hyperparameters in the two models can be tuned, like truncated mode, channel number and batch size. FNO can be implemented using the `NeuralOperator` package, which includes several common neural operator architectures. This package allows users to build models and use GPU acceleration directly, saving the effort of writing complex code like Listing 1. It makes the code cleaner and provides great convenience in comparing different operators on the same task. Additionally, the various loss functions in the package make it much simpler to build PINO by adding physical constraints to the model.

The internal architecture of the FNO offers a pathway for enhancing the explainability of the learned parameterization. As we mentioned in Section 2.2, the FNO shares structural similarities with CNNs. Previous studies utilizing CNNs for QBO parameterization have demonstrated that convolutional layers effectively function as multi-frequency filters and examined the Fourier spectra of kernels in each layer (Pahlavan et al. [2024]). The similar tests have the potential to be adopted on the FNO model here in order to check whether it performs like a filter. Furthermore, the FNO transcends the limitations of traditional CNNs through its discretization-invariance, which is a unique property of neural operators (Li et al. [2020]). This facilitates further exploration of QBO phase sensitivity to grid resolution, a topic that has been studied in conventional numerical simulations (Giorgetta et al. [2002]).

Finally, the QBO remains a challenging signal to faithfully reproduce in General Circulation Models (GCMs) (Garfinkel et al. [2022]). However, its accurate representation is very important, given its profound impact on a wide range of atmospheric phenomena and global teleconnections (Anstey et al. [2022b]). Bridging the gap from this idealized 1D testbed to embedding FNO-based parameterizations into complex 3D GCMs remains the ultimate task and still has a long way to go. Such integrations will be critical to verify whether the spectral advantages identified in this study can effectively mitigate systematic biases in tropical stratospheric modeling and enhance the overall fidelity of global climate predictions.

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